

Shankhanil Borthakur

shanknil@umich.edu | +1 (734) 210 7484 | [LinkedIn](#) | [GitHub](#) | [Website](#) | Open to internship (May-Aug 2027)

EDUCATION

University of Michigan, Ann Arbor

M.S. in Computer Science & Engineering

Aug 2026 - May 2028

GPA: 4/4

National Institute of Technology, Surat

B.Tech. in Computer Science (Minor), Electronics and Communication (Major) - Gold Medalist

Aug 2019 - May 2023

GPA: 3.95/4

TECHNICAL SKILLS

Programming: Python, C++, Java, JavaScript, SQL, HTML/CSS

Cloud & Data: AWS (Redshift, Aurora, Lambda, API Gateway, Kinesis, S3), Terraform, PostgreSQL, MySQL, MongoDB

ML / AI: PyTorch, TensorFlow/Keras, OpenCV, NumPy, Pandas, NLP, RAG & Agentic AI, OpenAI/Anthropic/Gemini APIs

Engineering: Django, Flask, Node.js, React, Docker, REST APIs, CI/CD, Git, Distributed Systems, Linux/Unix, Networks

Certifications: AWS Certified AI Practitioner [\[Credential\]](#), AWS Certified Cloud Practitioner [\[Credential\]](#)

PROFESSIONAL EXPERIENCE

Software Engineer II, JPMorgan Chase, India

Feb 2025 - Jun 2026

- Architected a **250 TB** Redshift data warehouse for a \$2B trading business; reduced annual storage costs by **\$150K** using hot/cold data tiering and Redshift Spectrum for on-demand S3 querying.
- Built a real-time pipeline to process **5M+ trades/day** with dual-write consistency across Redshift and Redis. Achieved **3X** write throughput over single-node INSERTs using Redshift COPY command.
- Led incident response for Redshift failover during 2 regional AWS outages, restoring service with **zero RPO** and **under 1 hour RTO**.
- Developed a **RAG agent** for AI-assisted analysis of Redshift distkey/sortkey configurations and vacuum/analyze schedules.

Software Engineer, JPMorgan Chase, India

Aug 2023 - Jan 2025

- Engineered MongoDB aggregation pipelines using **\$graphLookup** to traverse hierarchical time-series data. Built the backend for a dependency visualization dashboard that reduced cross-team Cron job delay MTTD from 2 hrs to 15 min.
- Built multi-tenant AWS API Gateway + Lambda endpoints to ingest Cron job telemetry into MongoDB with nested-document upserts.
- Recognized with an **accelerated promotion** – top 5% of the global early-career engineering cohort.

Software Engineer Intern, JPMorgan Chase, India

May 2022 - Jul 2022

- Designed a **Kinesis Firehose** pipeline from on-premise to S3 data lake. Configured dynamic partitioning to route incoming data to dedicated S3 bucket prefixes and boost analytical query performance.
- Built a serverless ETL job (Lambda + AWS Data Wrangler) to convert JSON payload to Parquet, cutting S3 storage footprint by **34%**.

PROJECTS

AI-Powered Virtual Fitness Platform [\[Code\]](#) [🔗](#)

- Developed a WebRTC application using **Peer.js** full-mesh architecture. Integrated **MediaPipe** pose detection model on HTML5 Canvas to extract 33 keypoints/frame for posture tracking.
- Designed a scale/translation-invariant embedder with a 2-stage KNN classifier. Integrated **Exponential Moving Average** (EMA) filter to smooth probability vectors across sequential frames and reduce prediction jitter.
- **1st place, HackeRam Hackathon – Fordham University, NYC (365 global participants)**

Smart Travel Planning using Named Entity Recognition [\[Code\]](#) [🔗](#)

- Built a **BERT-BiLSTM-CRF** model to extract and classify tourist attractions from travel descriptions. Integrated the model's output with Mapbox Directions API to plot an efficient route on a React frontend.
- Resolved word polysemy via 768-dim BERT embeddings and a bidirectional LSTM to encode forward/backward sequence context. Optimized sequence tagging via a linear-chain **CRF decoder**, achieving an **80.01%** F1-score.
- Published and presented the above research as first author at 2023 IEEE 5th ICAECC (IEEE Xplore-indexed) [\[Paper\]](#)
- **Sponsor track winner, Hack This Fall hackathon (506 participants, India-wide)**

RESEARCH WORK

Research Fellowship, IIT Delhi – Applied **Twin Neural Networks** to address class imbalance in a COVID-19 chest X-ray dataset having fewer positive samples, improving classification accuracy by **5%** over baseline CNN.

Indian Space Research Organization – Built a geolocation algorithm that maps **Mask R-CNN** instance segmentation outputs to GPS coordinates in satellite imagery, achieving sub-meter positional accuracy for remote sensing applications.